

Nan Hu

July 2026

CONTACT INFORMATION

Joint Institute for Food Safety and Applied Nutrition
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EDUCATION

Ph.D., Biology September 2017 – May 2023
Texas Tech University, Lubbock
Advisor: Matthew S. Olson, Ph.D.
Dissertation: Sex chromosome evolution in willows (*Salix* spp.)
M.S., Forest Genetics September 2014 – July 2017
Nanjing Forestry University, Nanjing
Advisor: Tongming Yin, Ph.D.
Thesis: Organelle genome assembly of *Salix wilsonii*
B.S., Forestry September 2010 – July 2014
Nanjing Forestry University, Nanjing
Thesis: Phenotypical variance of wood properties of a *Salix suchowensis* F1 offspring

RESEARCH POSITIONS

Post-doctoral Research Associate
University of Maryland/JIFSAN, College Park
Supervisor: Jianghong Meng, Ph.D (UMD) January 2025 – Present
Supervisor: Sara Handy, PhD (FDA)

Texas Tech University, Lubbock
Supervisor: Matthew G. Johnson, Ph.D August 2023 – December 2024
Program: Improving detection of plant contaminants in mixed samples
with targeted sequencing, genome skimming, and population genetics

GRANT AND FUNDING

Texas Tech University Graduate Research Support Funding -(\$1,000)
PI, Awarded. Does sex allocation result in clonal size differences in the dioecious plant *Salix exigua*? (2019-2020)

USDA-NIFA The Agriculture and Food Research Initiative (AFRI)
(\$649,923)
Co-PI, Pending. Genomic tools for plant authentication, geographic origin, and heavy metal risk verification in botanical supply chains (2026-2030)

DARPA (Defense Advanced Research Projects Agency) AG X BTO (up to \$500,000)
Co-PI, Pending. Global plant SNP atlas-A universal genomic tools to assess provenance and human intervention (2026-2027)

FDA Broad Agency Announcement (\$369,698)
Co-PI, Pending. Genomic tools for plant authentication, geographic origin, and heavy metal risk verification in botanical supply chains (2026-2029)

PUBLICATIONS

1. **Hu, N.**, Bullock, M., Jackson, C., Miller C., Hunter, E., Huff, C., Handy, S., Johnson, M. SPrOUT: A computational and targeted sequencing approach for mixed plant DNA identification with Angiosperms353. *Applications in Plant Sciences*. In press. <https://doi.org/10.64898/2026.02.20.707031>
2. Khanal, A., **Hu, N.**, Guo, M., Gambhir, D., and Olson, M. S (2025). Low reproductive output suggests hermaphrodites are unlikely to spread in a dioecious hybrid willow population. *International Journal of Plant Sciences* <https://doi.org/10.1086/738603>
3. Jiao, Y., **Hu, N.**, Li, J., Xia, H., Li, X., Liu, Y., Wang, J. (2025) Multi-omics approaches investigate the bitter flavor in the shoot of *Bambusa oldhamii* Munro. *BMC Plant Biol* **25**, 1501. <https://doi.org/10.1186/s12870-025-07529-z>
4. Guo, M., **Hu, N.**, Feng, G., Wang, L., Gambhir, D., Khanal, A., Hyden, B., Smart, L.B., Sanderson, B.J. and Olson, M.S. (2025), Divergence, hybridization, and diversification in an eastern North American willow syngameon. *New Phytologist*. <https://doi.org/10.1111/nph.70469>
5. Khanal, A., **Hu, N.**, Guo, M., Gambhir, D., Sanderson, B., and Olson, M. S (2025). Rare recombination in *Salix nivalis* and the maintenance of homomorphic sex chromosomes in willows. *American Journal of Botany*, e70059. <https://doi.org/10.1002/ajb2.70059>
6. Gambhir, D., Sanderson, B. J., Guo, M., **Hu, N.**, Khanal, A., Cronk, Q., and Olson, M. S. (2025). Disentangling serial chloroplast captures in willows. *American Journal of Botany*, 112(5), e70039. <https://doi.org/10.1002/ajb2.70039>
7. **Hu, N.**, Sanderson, B. J., Guo, M., Feng, G., Gambhir, D., Hale, H., Wang, D., Hyden, B., Liu, J., Smart, L. B., DiFazio, S. P., Ma, T., and Olson, M. S. (2023). Evolution of a ZW sex chromosome system in willows. *Nature Communications*, 14(1), 7144. <https://doi.org/10.1038/s41467-023-42880-5>
8. Sanderson, B. J., Gambhir, D., Feng, G. Q., **Hu, N.**, Cronk, Q. C., Percy, D. M., Freaner, F. M., Johnson, M. G., Smart, L. B., Keefover-Ring, K., Yin, T. M., Ma, T., DiFazio, S. P., Liu, J. Q., and Olson, M. S. (2023). Phylogenomics reveals patterns of ancient hybridization and differential diversification that contribute to phylogenetic conflict in willows, poplars, and close relatives. *Systematic Biology*, 72(6), 1220-1232. <https://doi.org/10.1093/sysbio/syad042>
9. **Hu, N.**, Hale, H., Sanderson, B., Feng, G. Q., Guo, M. H., Gambhir, D., & Olson, M. (2023). Reproductive Sexual Dimorphisms in Two Willow Species, *Salix exigua* Nutt. and *Salix nigra* Marshall. *International Journal of Plant Sciences*, 184(9), 688-695. <https://doi.org/10.1086/726623>
10. Liu, H. L., Wang, X. B., Wang, G. B., Cui, P., Wu, S. G., Ai, C., **Hu, N.**, Li, A. L., He, B., Shao, X. J., Wu, Z. C., Feng, H., Chang, Y. X., Mu, D. S., Hou, J., Dai, X. G., Yin, T. M., Ruan, J., & Cao, F. L. (2021). The nearly complete genome of *Ginkgo biloba* illuminates gymnosperm evolution. *Nature Plants*, 7(6), 748-+. <https://doi.org/10.1038/s41477-021-00933-x>
11. Sanderson, B. J., Feng, G. Q., **Hu, N.**, Carlson, C. H., Smart, L. B., Keefover-Ring, K., Yin, T. M., Ma, T., Liu, J. Q., DiFazio, S. P., & Olson, M. S. (2021). Sex determination through X–Y heterogamety in *Salix nigra*. *Heredity*, 126(4), 630-639. <https://doi.org/10.1038/s41437-020-00397-3>
12. Chen, Y.[‡], **Hu, N.**[‡], & Wu, H. (2019). Analyzing and characterizing the chloroplast genome of *Salix wilsonii*. *Biomed Research International*, 2019, 5190425. <https://doi.org/10.1155/2019/5190425>

13. Olson, M. S., Sanderson, B., **Hu, N.**, Feng, G., & DiFazio, S. (2019). Diverse Sex Chromosomes in the Salicaceae. Plant and Animal Genome XXVII Conference (January 12-16, 2019),
14. Liu, H. L., Yang, W. X., Hou, J., **Hu, N.**, Yin, T. M., & Li, S. X. (2016). Genetic identification of 43 elite clonal accessions of *Populus deltoides* by SSR fingerprinting. *Canadian Journal of Plant Science*, 96(3), 494-502. <https://doi.org/10.1139/cjps-2015-0272>
15. Zhou, W., Tang, Z., Hou, J., **Hu, N.**, & Yin, T. (2015). Genetic Map Construction and Detection of Genetic Loci Underlying Segregation Distortion in an Intraspecific Cross of *Populus deltoides*. *Plos One*, 10(5), e0126077. <https://doi.org/10.1371/journal.pone.0126077>

‡: Co-first authorship

MANUSCRIPTS IN REVIEW/PREP

1. **Hu, N.**, Sanderson, B. J., Guo, M., Feng, G., Gambhir, D., Hale, H., Wang, D., Hyden, B., DiFazio, S. P., Olson, M. S. Shared SNP loci reveal the ancestor sex chromosomes in the Salix clade. In prep. To be submitted in Oct. 2026
2. **Hu, N.**, Hunter, E., Litterman, R., Wen, J., Ott, B., Greenbaum A., and Handy, S. A Vouchered Chloroplast Genome Resource and Diagnostic Marker-Discovery Framework for Botanical Identification. In review.
3. Cai, Y.‡, **Hu, N.**‡, Ren, Y., Wang, J., Yang, T., Yang, X., Kong, X., Yang Y., and Yang Y. Chromosome-scale Euphorbia genomes reveal repeat-associated genome evolution and locus variation at terpene synthase genes in Euphorbiaceae. In review.
4. **Hu, N.**, Bullock, M., Johnson, M., Micallef, S., Handy, S., and Choo, J. Beyond DNA barcoding: A hierarchical strategy for botanical material identification. In prep. To be submitted in Sept. 2026

‡: Co-first authorship

TEACHING AND MENTORING

Lecture/Workshop:

Texas Tech University

Target sequencing workshop

Spring 2022

- 6 sessions in fulfilling the increasing needs in target sequencing
- Bioinformatics lecture for both beginners and experienced
- Introduced target sequencing data preparation and downstream analysis
- Designed and lectured on session 05: Species Tree using target sequencing

Course Coordinator:

Plant Biology

Spring 2019, Spring 2020

- Lab coordinator in Spring 2019 and Spring 2020 for curriculum design and lab supplies preparation

TEACHING
AND
MENTORING

Teaching Assistant:

Department of Biological Sciences, Texas Tech University

Biology I/II

Spring 2022 – Spring 2023

- General college core biology course. Topics cover basic botany, entomology, anatomy, microbiology, genetics, ecology, and biochemistry
- Associated credit labs in full 3 hours, two sections a week
- Each lab is a combination of 1-1.5 hours lecturing and 1.5-2 hours of wet labs
- 4 sets of quizzes designed weekly
- 2 group projects per semester
- The teaching is monitored and practiced under the guidance of biological pedagogy

Plant Biology (Major)

Fall 2021

- Plant biology/botany course for majors
- Guided a weekly field sampling and identification session
- Guided an iNaturalist collection project

Population Genetics (Graduate Level)

Spring 2021

- Molecular population genetics for graduate students
- Preparing weekly projects materials
- Leading discussions about research questions
- Developed computational pipelines for practical studies of molecular population genetics

Organic Evolution

Fall 2019, Fall 2020

- Core course of biology major
- Hosted discussion sessions about both macroevolution and microevolution topics
- Assisting students for poster presentations on various evolutionary topics

Plant Biology (Non-major)

Spring 2018– Spring 2019, Spring 2020

- Non-major plant biology course including understanding basic structure and physiology of plants
- 1-credit labs training for scientific methods and basic lab skills in plant biology
- 2 projects per session

Genetics

Fall 2017

- Core course of biology major
- Leading the review sessions
- Including 18 hours of wet labs

TEACHING
AND
MENTORING

Undergraduate and Graduate Students Mentoring:

Graduate Student: Oluwaseun Shodipo (2023-Now)

Bioinformatics in phylogenetic inferences of *Haplopappus*. We looked into how fast speciation happened along with geographic distributions in North and South Americas

Undergraduate Students: Courtney Miller (2023); Charles Huff (2024-2025)

Generate testing mixed samples for mixed plant DNA identification pipeline. Presented in CUREs

Undergraduate Students: Sydney Terry (2020); Manjita Sharma (2020)

Extract SNP information from population genetics datasets to design genetic markers for identifying individual plants from stem cuttings. Presented in Undergraduates Poster Presentation Competition

CURRICULUM
DESIGN

Advanced Molecular Population Genetics

Spring 2021

<https://github.com/nhu92/popgen2021>

Full set of molecular population genetics practices in Bash and R, using real large datasets

Target sequencing workshop using Angiosperms353

Spring 2022

<https://github.com/ELReedHerbarium/TargetCaptureWorkshop>

Introduced target sequencing data preparation and downstream analysis

CONFERENCE
SYMPOSIA

Botany 2026 – Biodiversity at the Boundaries, Tucson, Arizona

Invited Speaker: From Phylogenomics to Population Genomics: Assessing Angiosperms353 for Fine-Scale Genetic Resolution

IAFP 2026 Annual Meeting, New Orleans, Louisiana

Poster Presentation: An Expanded FDA Botanical DNA Reference Library and Marker Evaluation Framework for Plant Species Identification

Poster Presentation: SPrOUT: A computational and targeted sequencing approach for mixed plant DNA identification with Angiosperms353

Botany 2025 – Botany without Barriers, Palm Springs, California

Presentation: A computational and targeted sequencing approach for mixed plant DNA identification with Angiosperms353

International Botanical Congress, Madrid-the 20th IBC

Poster Presentation: Mixed plant DNA identification with Angiosperms353: A computational and targeted sequencing approach

Texas Tech Graduate School Poster Presentation Competition

Poster Presentation: Mapping the sex determination region in dioecious species *Salix exigua*

Texas Tech Annual Biological Sciences Symposium

Presentation: Sex determination in narrow-leaf willow (*Salix exigua*)

AWARDS

Biology Teaching Scholar

Texas Tech University

Certified Biology Teaching Scholar awarded through strict biology pedagogy evaluations

Graduate School Poster Presentation Competition

Texas Tech University

First Prize in Plant Science

Title: Mapping the sex determination region in dioecious species *Salix exigua*

Graduate Research Support Funding

Texas Tech University

Proposal title: Does sex allocation result in clonal size differences in the dioecious plant *Salix exigua*?

PROFESSIONAL
SERVICES

Contributed Peer Review: *New Phytologist, American Journal of Botany, Plants, Biology, Forest science and technology*

Workshop Coordinator of West Texas Association of Biology

Judge of Texas Tech Annual Biological Sciences Symposium

Judge of Texas Tech Poster Presentation Competition